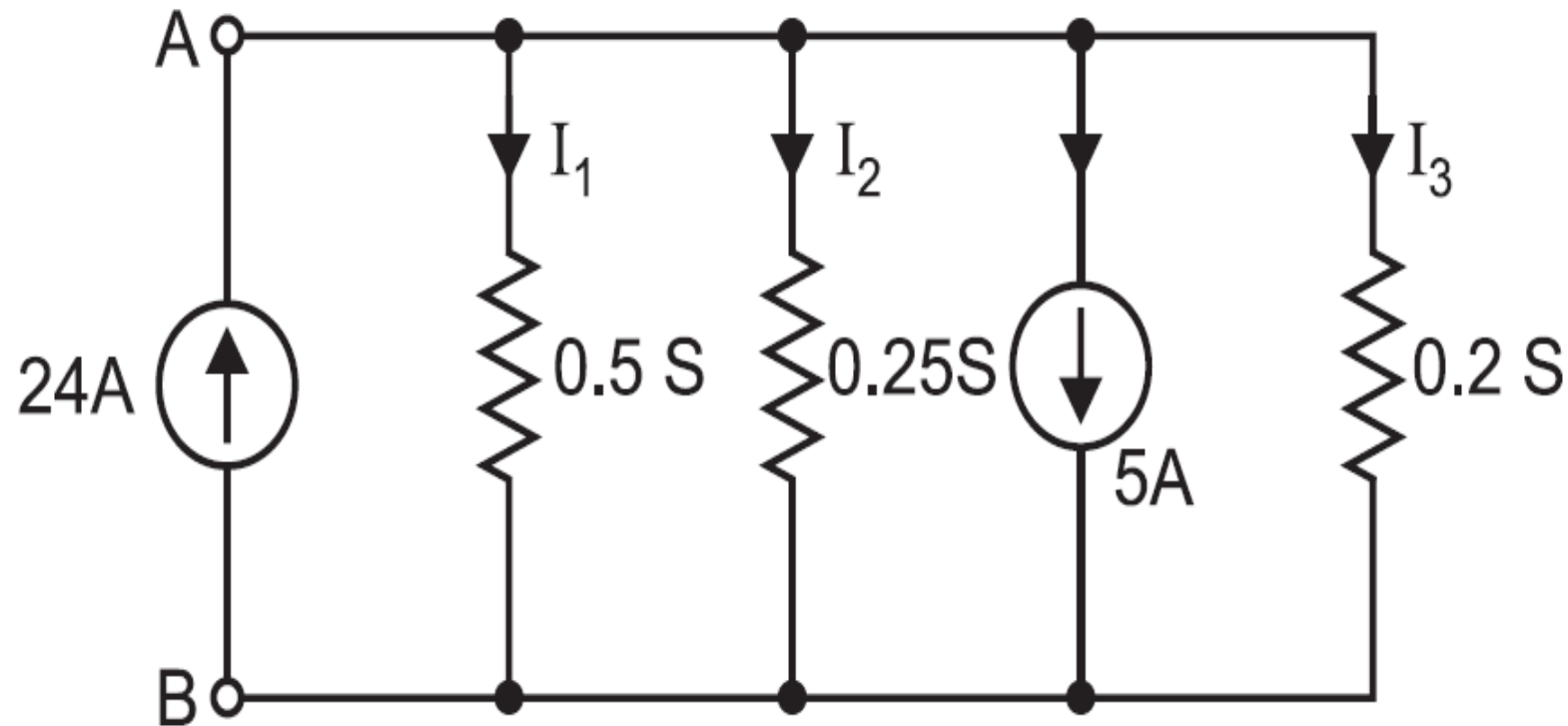




EXERCISE/NUMERICAL ANALYSIS



Q. Find the three branch currents in the circuit shown below. What is the potential difference between points A and B?





EXERCISE/NUMERICAL ANALYSIS



Soln: Current sources in parallel add algebraically.

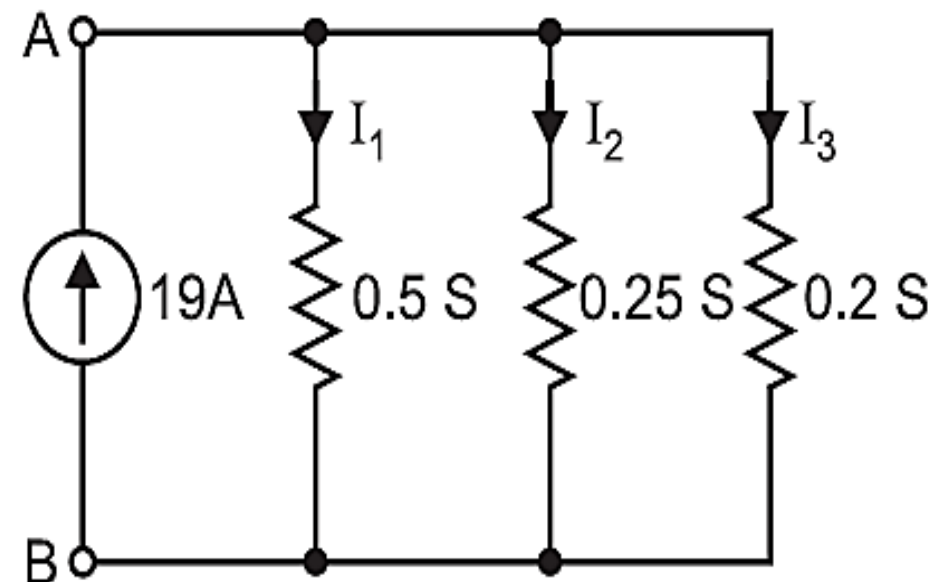
Therefore, the two current sources can be combined to give the resultant current source of current, $I = 24 - 5 = 19 \text{ A}$

Total circuit conductance is

$$\begin{aligned} G_T &= G_1 + G_2 + G_3 \\ &= .5 + 0.25 + 0.2 = 0.95 \text{ S} \end{aligned}$$

Applying Current Division Rule,

$$I_1 = \frac{G_1}{G_p} I = \frac{0.5}{0.95} \times 19 = 10 \text{ A}$$





EXERCISE/NUMERICAL ANALYSIS



$$I_2 = \frac{G_2}{G_p} I = \frac{0.25}{0.95} \times 19 = 5 \text{ A}$$

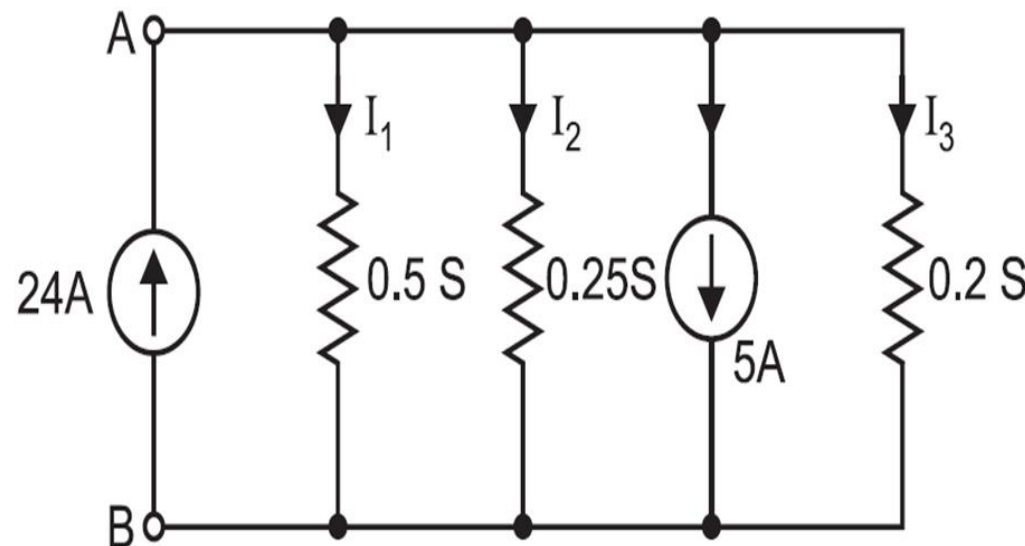
$$I_3 = \frac{G_3}{G_p} I = \frac{0.2}{0.95} \times 19 = 4 \text{ A}$$

As all branches are in parallel, voltage across each branch is same.

Hence,

$$V_{AB} = \frac{I_1}{G_1} = \frac{I_2}{G_2} = \frac{I_3}{G_3}$$

$$V_{AB} = \frac{10}{0.5} = \frac{5}{0.25} = \frac{4}{0.2} = 20 \text{ V}$$

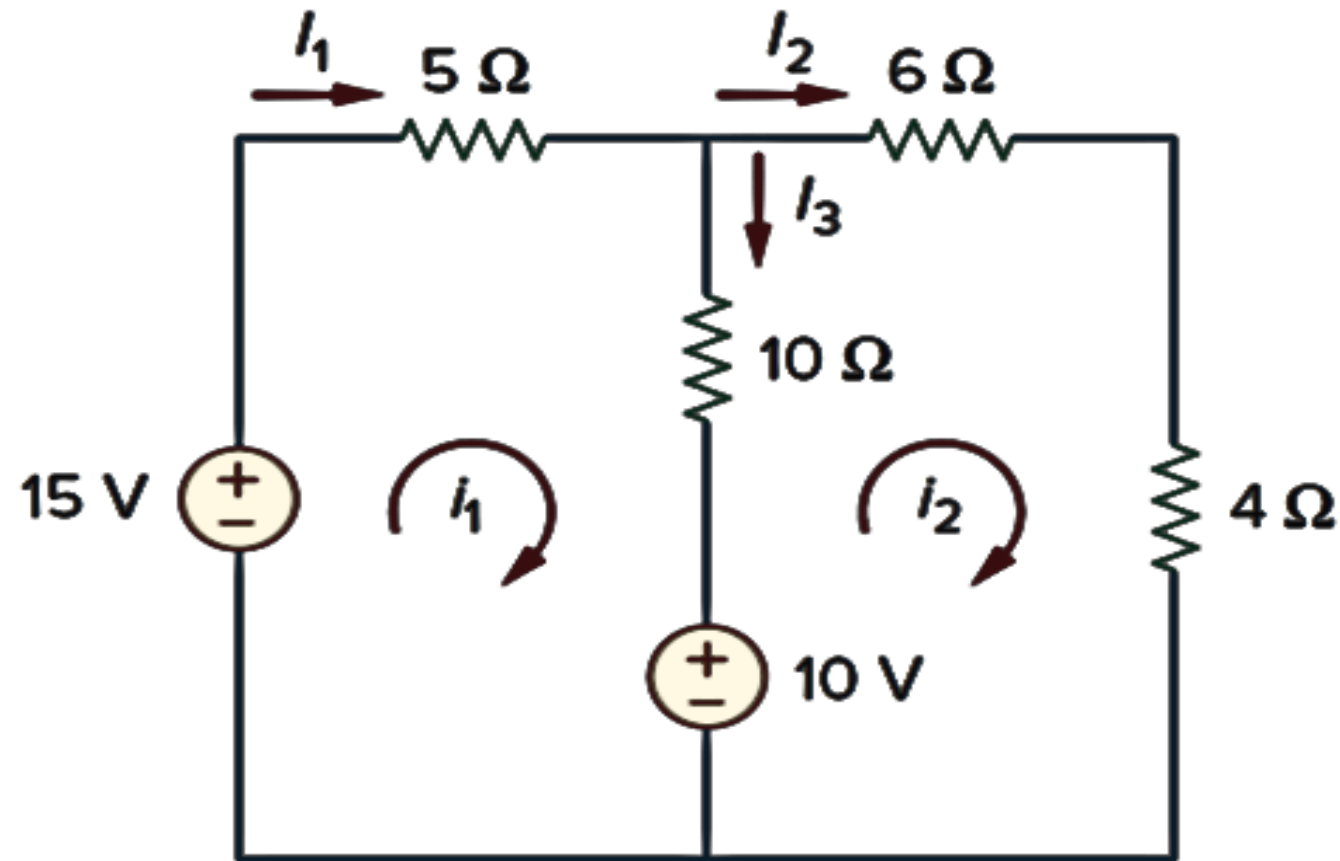




EXERCISE/NUMERICAL ANALYSIS



Q. Find mesh currents i_1 , i_2 , i_3 using mesh analysis.





EXERCISE/NUMERICAL ANALYSIS



Applying KVL in mesh 1.

$$-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$$

$$\Rightarrow 3i_1 - 2i_2 = 1 \quad (1)$$

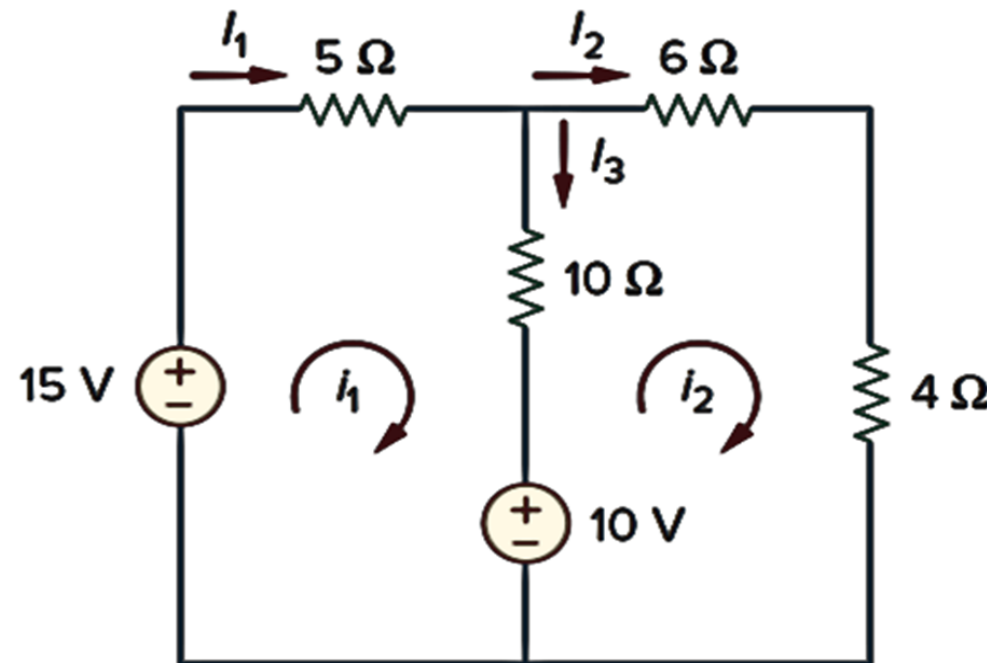
Applying KVL in mesh 2,

$$6i_2 + 4i_2 + 10(i_2 - i_1) - 10 = 0$$

$$\Rightarrow i_1 - 2i_2 = -1 \quad (2)$$

Solving above two equations we have

$$i_1 = 1A, i_2 = 1A, i_3 = (i_1 - i_2) = 0A$$





EXERCISE/NUMERICAL ANALYSIS



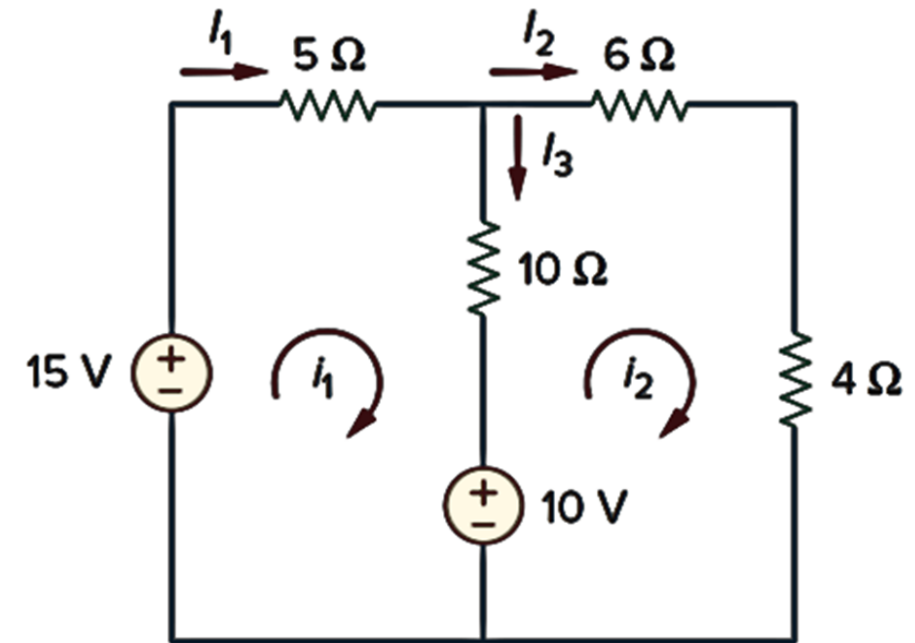
Eqs. (1) and (2) can also be solved using Cramer's rule.

In matrix form we obtain,

$$\begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Calculating determinants,

$$\Delta = \det \begin{bmatrix} 3 & -2 \\ -1 & 2 \end{bmatrix} = 6 - 2 = 4$$





EXERCISE/NUMERICAL ANALYSIS

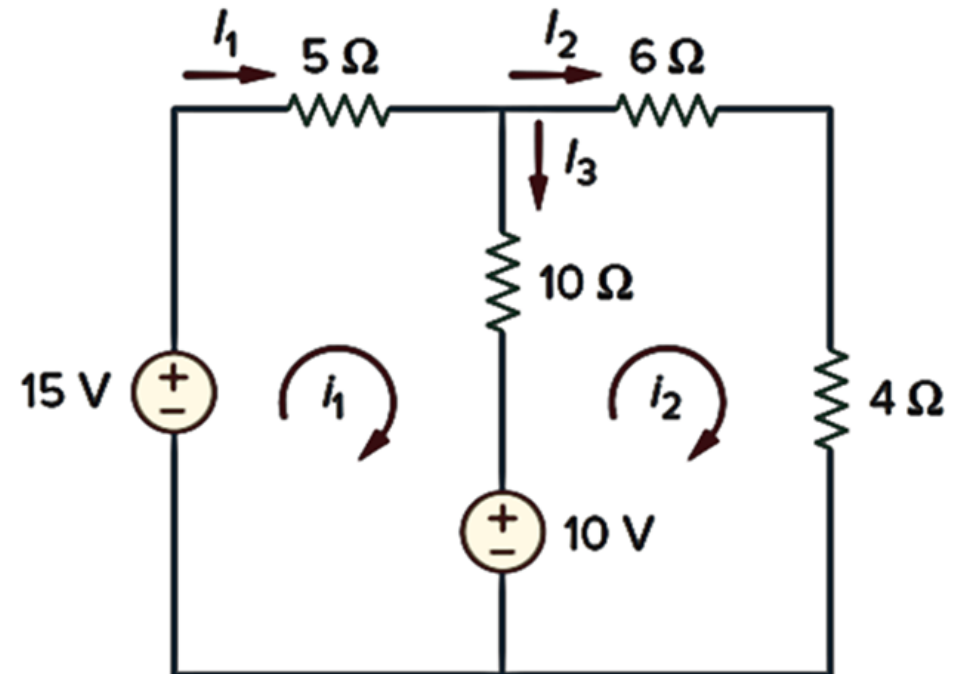


$$\Delta_1 = \det \begin{bmatrix} 1 & -2 \\ 1 & 2 \end{bmatrix} = 2 + 2 = 4$$

$$\Delta_2 = \det \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix} = 3 + 1 = 4$$

$$i_1 = \frac{\Delta_1}{\Delta} = 1A, i_2 = \frac{\Delta_2}{\Delta} = 1A$$

$$i_3 = i_1 - i_2 = 0A$$

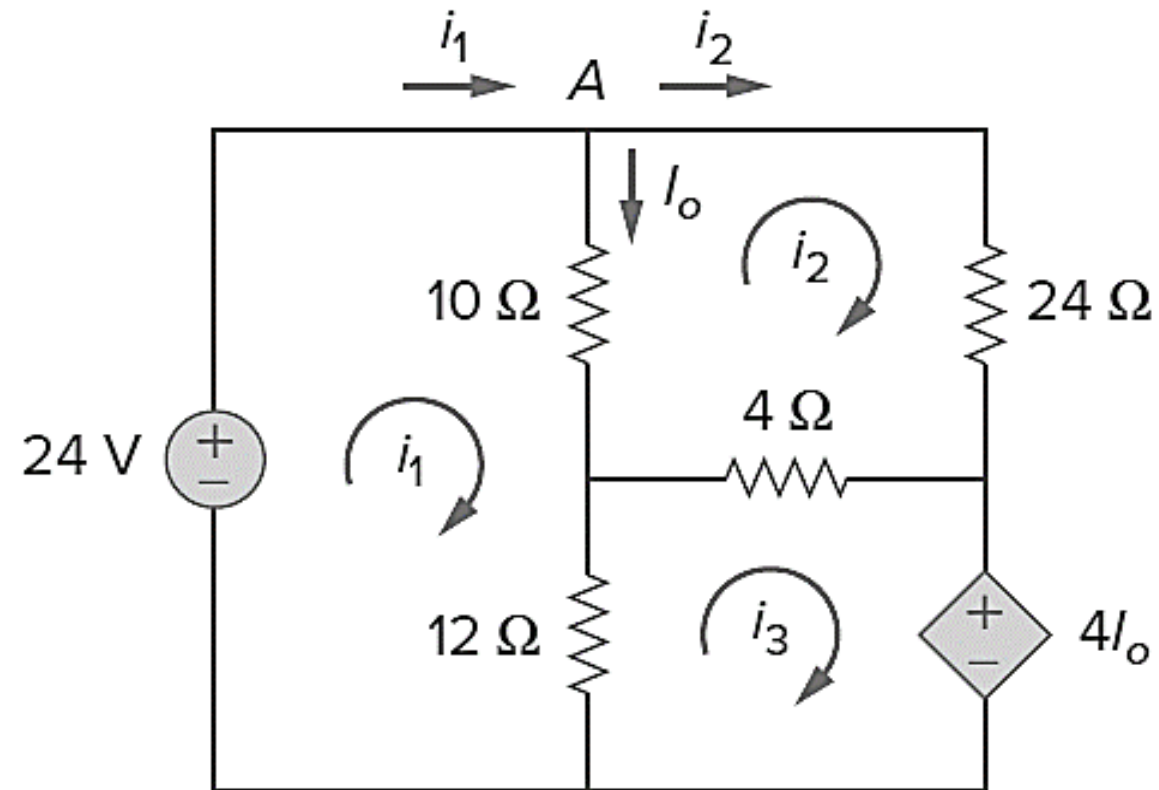




EXERCISE/NUMERICAL ANALYSIS



Q. Find mesh currents i_1 , i_2 , i_3 and i_o using mesh analysis.





EXERCISE/NUMERICAL ANALYSIS



Applying KVL in mesh 1.

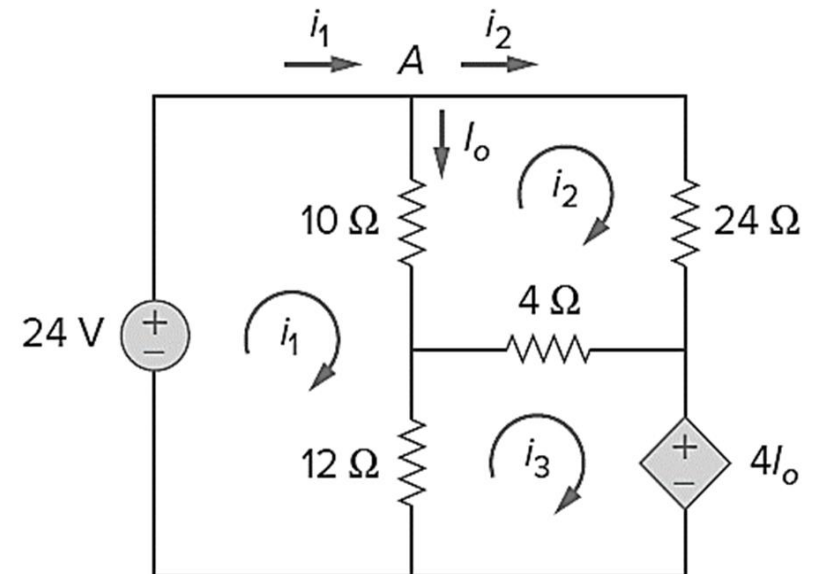
$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0 \Rightarrow 11i_1 - 5i_2 - 6i_3 = 12 \quad (1)$$

Applying KVL in mesh 2,

$$24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0 \Rightarrow -5i_1 + 19i_2 - 2i_3 = 0 \quad (2)$$

Applying KVL in mesh 3

$$4I_o + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$





EXERCISE/NUMERICAL ANALYSIS



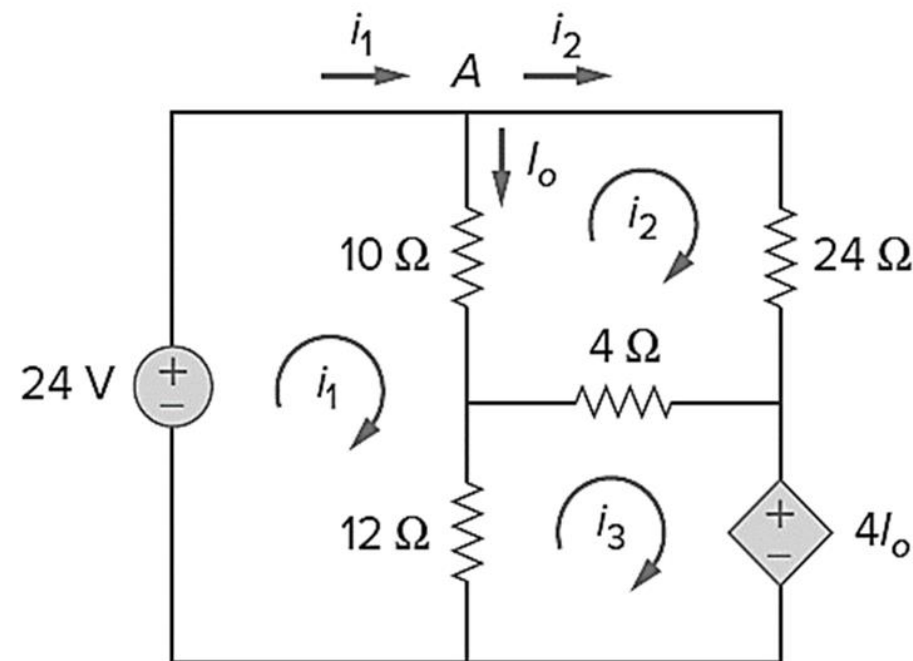
But at node A, $I_o = i_1 - i_2$, so that

$$\Rightarrow 4(i_1 - i_2) + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

$$\Rightarrow -i_1 - i_2 + 2i_3 = 0 \quad (3)$$

Using cramer's' rule to solve 3 unknown variables and writing in matrix form.

$$\begin{bmatrix} 11 & -5 & -6 \\ -5 & 19 & -2 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$$





EXERCISE/NUMERICAL ANALYSIS

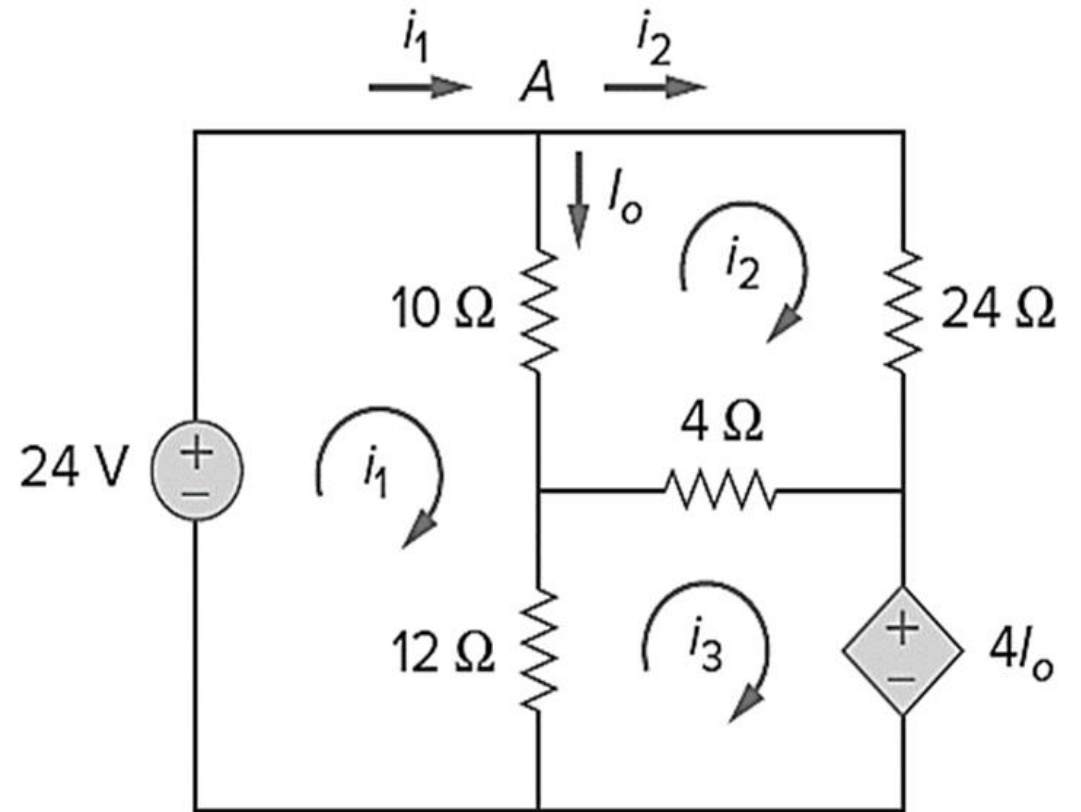


$$\Delta = \det \begin{bmatrix} 11 & -5 & -6 \\ -5 & 19 & -2 \\ -1 & -1 & 2 \end{bmatrix} = 192$$

$$\Delta_1 = \det \begin{bmatrix} 12 & -5 & -6 \\ 0 & 19 & -2 \\ 0 & -1 & 2 \end{bmatrix} = 432$$

$$\Delta_2 = \det \begin{bmatrix} 11 & 12 & -6 \\ -5 & 0 & -2 \\ -1 & 0 & 2 \end{bmatrix} = 144$$

$$\Delta_3 = \det \begin{bmatrix} 11 & -5 & 12 \\ -5 & 19 & 0 \\ -1 & -1 & 0 \end{bmatrix} = 288$$

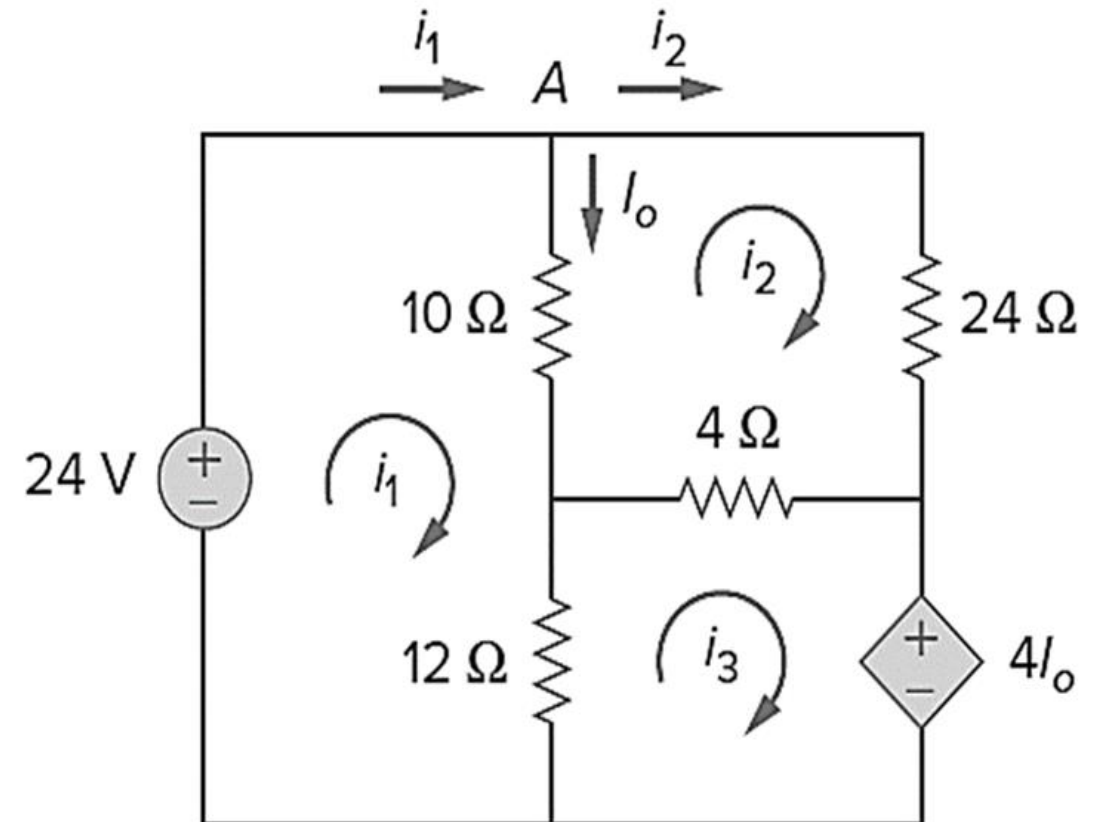




EXERCISE/NUMERICAL ANALYSIS



$$i_1 = \frac{\Delta_1}{\Delta} = 2.25A, i_2 = \frac{\Delta_2}{\Delta} = 0.75A$$
$$i_3 = \frac{\Delta_3}{\Delta} = 1.5A$$

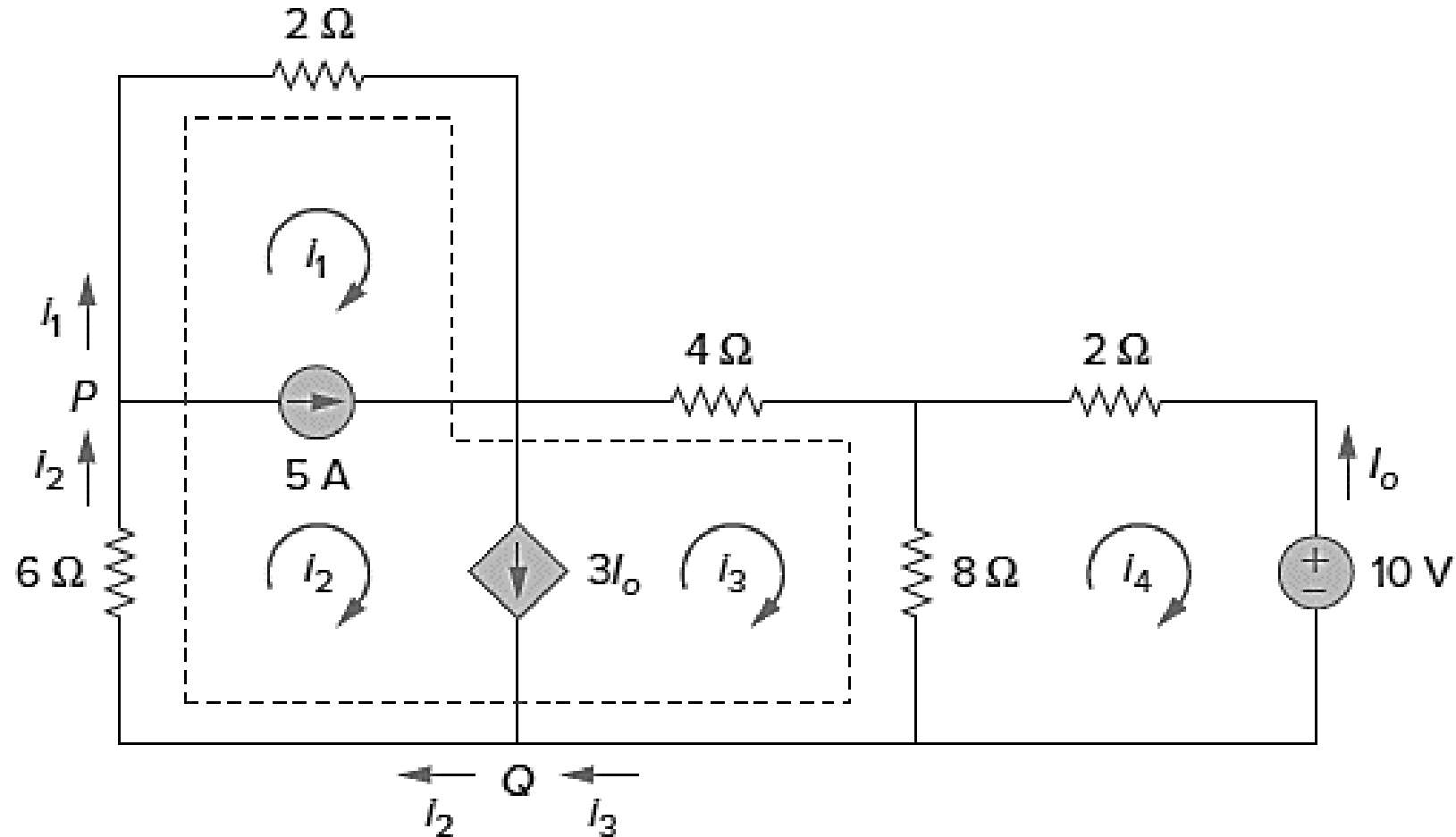




EXERCISE/NUMERICAL ANALYSIS



Q. Find mesh currents i_1 , i_2 , i_3 and i_4 using mesh analysis.





EXERCISE/NUMERICAL ANALYSIS



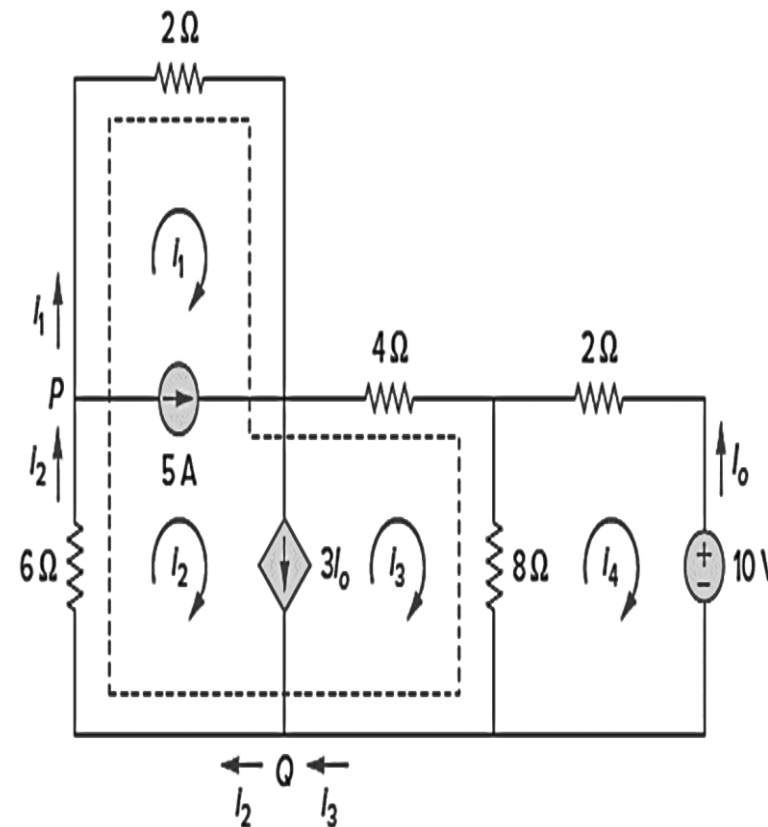
Soln:

Note : meshes 1 and 2 form a supermesh because they have an independent current source in common.

Note : meshes 2 and 3 form another supermesh because they have a dependent current source in common.

The two supermeshes intersect and form a larger supermesh as shown. Applying KVL to the larger supermesh,

$$2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 = 0$$





EXERCISE/NUMERICAL ANALYSIS



$$\Rightarrow i_1 + 3i_2 + 6i_3 - 4i_4 = 0 \quad \dots(1)$$

Applying KCL to node P: $i_2 = i_1 + 5 \quad \dots(2)$

Applying KCL to node Q: $i_2 = i_3 + 3I_o$

but $I_o = -i_4$

Hence, $i_2 = i_3 - 3i_4 \quad \dots(3)$

Applying KVL in mesh 4,

$$\Rightarrow 2i_4 + 8(i_4 - i_3) + 10 = 0 \quad \dots(4)$$

Solving (1),(2),(3) and (4) we obtain

$$i_1 = -7.5A, i_2 = -2.5A, i_3 = 3.93A \text{ and } i_4 = 2.143A$$

